

Project Report Template

Advanced Analytics and Applications

Team 02



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1 Problem description

The smart mobility market is highly competitive and current reports indicate that particularly ride-hailing, using autonomous and electric vehicles will dominate the market in the near future. Traditional car manufacturers face the risk of disruption if they fail to establish competitive mobility services. Our client, a renowned German car manufacturer, seeks to enter the ride-hailing market in the US, using a fully electrified fleet. However, the client lacks operational expertise in managing electric vehicle fleets in an urban ride-hailing context. Key challenges include understanding spatial and temporal demand patterns to deploy vehicles efficiently and optimizing charging schedules to reduce energy costs while ensuring vehicles are always sufficiently charged for their working day.

To address these challenges, we analyze historical taxi trip data from Chicago as a proxy for ride-hailing demand. Specifically, we aim to conduct a descriptive analysis, which analyzes taxi demand patterns and how they vary in different spatio-temporal resolutions. Second, we build predictive models that forecast taxi trip demand for a given spatial unit and time window, enabling the client to proactively position vehicles where and when they are needed. For example, this includes estimating expected demand on a rainy Sunday morning in the city outskirts. Furthermore, we develop an intelligent charging strategy using reinforcement learning to minimize charging costs while ensuring sufficient vehicle range. Together, these analyses will enable the client to make data-driven decisions regarding fleet deployment, driver scheduling, and charging infrastructure investment.

2 Data description

To approach this task, data from multiple sources was used. The primary dataset used in this analysis is the Chicago Taxi Trips dataset (City of Chicago, 2024), which is publicly available via the Chicago Data Portal and contains taxi trips from 2024 onwards. The Taxi Trip Dataset contains one trip per row, with timestamps rounded to the nearest 15 minutes. Since the full dataset starting from 2024 contains over 16 million rows, the dataset was limited to only include trips from 2025, resulting in a raw dataset of roughly 6.8 million rows. Important attributes include trip and taxi identifiers, the timestamp of the trip, centroid coordinates of the community area or census tract where the trip took place and multiple attributes about the trips properties like price or length (Figure 1).

Furthermore, hourly Chicago weather data was obtained from the Open-Meteo API (Open-Meteo, 2026) for the period of January 2024 to January 2026. The initial dataset includes roughly 17500 entries and key variables of the dataset include temperature, precipitation and sunshine duration (Figure 2).

Points-of-Interest data for Chicago was gathered from OpenStreetMap via the Overpass API (OpenStreetMap Contributors, 2026), covering nine categories: airports, train stations, stadiums, restaurants, bars and clubs, hotels, hospitals, universities, and tourist attractions. These raw POI locations are subsequently aggregated into distance and density features for multiple spatial resolutions during data preparation.

In order to be able to analyse the data on census tract and community area level, additional geometry data containing the shapes of these areas was obtained from . These datasets contain the geometry for Chicago's 77 community areas and 878 census tracts.

3 Data Preparation

Before analysis, each dataset underwent a series of preparation steps to ensure data quality and compatibility across sources.

3.1 Trip Data

The dataset was first inspected to get an overview of the datatypes and find potential issues. This revealed that several numeric columns were incorrectly stored as strings and that the timestamp columns were not yet in datetime format. Both were corrected during preprocessing. Additionally, the dataset contained no duplicated rows or trip identifiers. Furthermore, several implausible trips were identified and removed during cleaning. These included trips with zero or negative distance (528447 trips) and trips with a duration of less than 60 seconds (381488 trips), as these trips are likely erroneous. Overlapping trips with the same taxi identifier were also removed (28320 trips) based on the assumption that these result from recording errors. Price attributes were also validated by checking for and removing trips with zero or negative fares (1455 trips) and whether there are negative tips, tolls or extras (0 trips).

The dataset was also checked for missing values, revealing that 0.18% of the non-spatial data was missing. Due to the negligible share, these rows were dropped. Because of privacy reasons, the spatial data showed many more missing values with census tract data missing for approximately 55% of trips. In these cases, the less granular community area is recorded instead, meaning spatial information is always available at either the census tract or community area level [TRUE ???].

The numeric columns were checked for outliers using boxplots (Figure 3) and removed using the z-score as a statistical solution. Observations with a z-score exceeding a threshold of 2 in any numeric column were considered outliers and removed. [REASON]. This resulted in 444102 rows, or 7.18% of the data at that point being removed, which was seen as acceptable since the remaining set still contained roughly 5.7 million rows.

[CONTINUE]

3.2 Supplementary Data

4 Data Analytics

test4

4.1 descriptive analysis

test

4.2 Analysis using SVM

test

4.3 Analysis using Neural Networks

test

4.4 Analysis using Reinforcement Learning

test

5 Conclusion

conclusion text

6 Appendix

Columns (23)			
Column Name	Description	API Field Name	Data Type
 Trip ID	A unique identifier for the trip.	trip_id	Text
 Taxi ID	A unique identifier for the taxi.	taxi_id	Text
 Trip Start Timestamp	When the trip started, rounded to the nearest 15 minutes.	trip_start_timestamp	Floating Timestamp
 Trip End Timestamp	When the trip ended, rounded to the nearest 15 minutes.	trip_end_timestamp	Floating Timestamp
 Trip Seconds	Time of the trip in seconds.	trip_seconds	Number
 Trip Miles	Distance of the trip in miles.	trip_miles	Number
 Pickup Census Tract	The Census Tract where the trip began. For privacy, this Census Tract is not shown for some trips. This column often will be blank for locations outside Chicago.	pickup_census_tract	Text
 Dropoff Census Tract	The Census Tract where the trip ended. For privacy, this Census Tract is not shown for some trips. This column often will be blank for locations outside Chicago.	dropoff_census_tract	Text
 Pickup Community Area	The Community Area where the trip began. This column will be blank for locations outside Chicago.	pickup_community_area	Number
 Dropoff Community Area	The Community Area where the trip ended. This column will be blank for locations outside Chicago.	dropoff_community_area	Number
 Fare	The fare for the trip.	fare	Number
 Tips	The tip for the trip. Cash tips generally will not be recorded.	tips	Number
 Tolls	The tolls for the trip.	tolls	Number
 Extras	Extra charges for the trip.	extras	Number
 Trip Total	Total cost of the trip, the total of the previous columns.	trip_total	Number
 Payment Type	Type of payment for the trip.	payment_type	Text
 Company	The taxi company.	company	Text
 Pickup Centroid Latitude	The latitude of the center of the pickup census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	pickup_centroid_latitude	Number
 Pickup Centroid Longitude	The longitude of the center of the pickup census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	pickup_centroid_longitude	Number
 Pickup Centroid Location	The location of the center of the pickup census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	pickup_centroid_location	Point
 Dropoff Centroid Latitude	The latitude of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	dropoff_centroid_latitude	Number
 Dropoff Centroid Longitude	The longitude of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	dropoff_centroid_longitude	Number
 Dropoff Centroid Location	The location of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	dropoff_centroid_location	Point

Figure 1: Raw Chicago Taxi Trip Data Column Descriptions

```

RangeIndex: 17568 entries, 0 to 17567
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   date                                  17568 non-null  datetime64[ms, UTC]
1   temperature_2m                        17568 non-null  float32
2   relative_humidity_2m                 17568 non-null  float32
3   apparent_temperature                 17568 non-null  float32
4   precipitation                         17568 non-null  float32
5   rain                                 17568 non-null  float32
6   snowfall                             17568 non-null  float32
7   snow_depth                           17568 non-null  float32
8   surface_pressure                     17568 non-null  float32
9   cloud_cover                          17568 non-null  float32
10  wind_speed_10m                       17568 non-null  float32
11  wind_speed_100m                      17568 non-null  float32
12  is_day                               17568 non-null  float32
13  sunshine_duration                    17568 non-null  float32
14  direct_radiation                     17568 non-null  float32
dtypes: datetime64[ms, UTC](1), float32(14)

```

Figure 2: Weather Data Columns

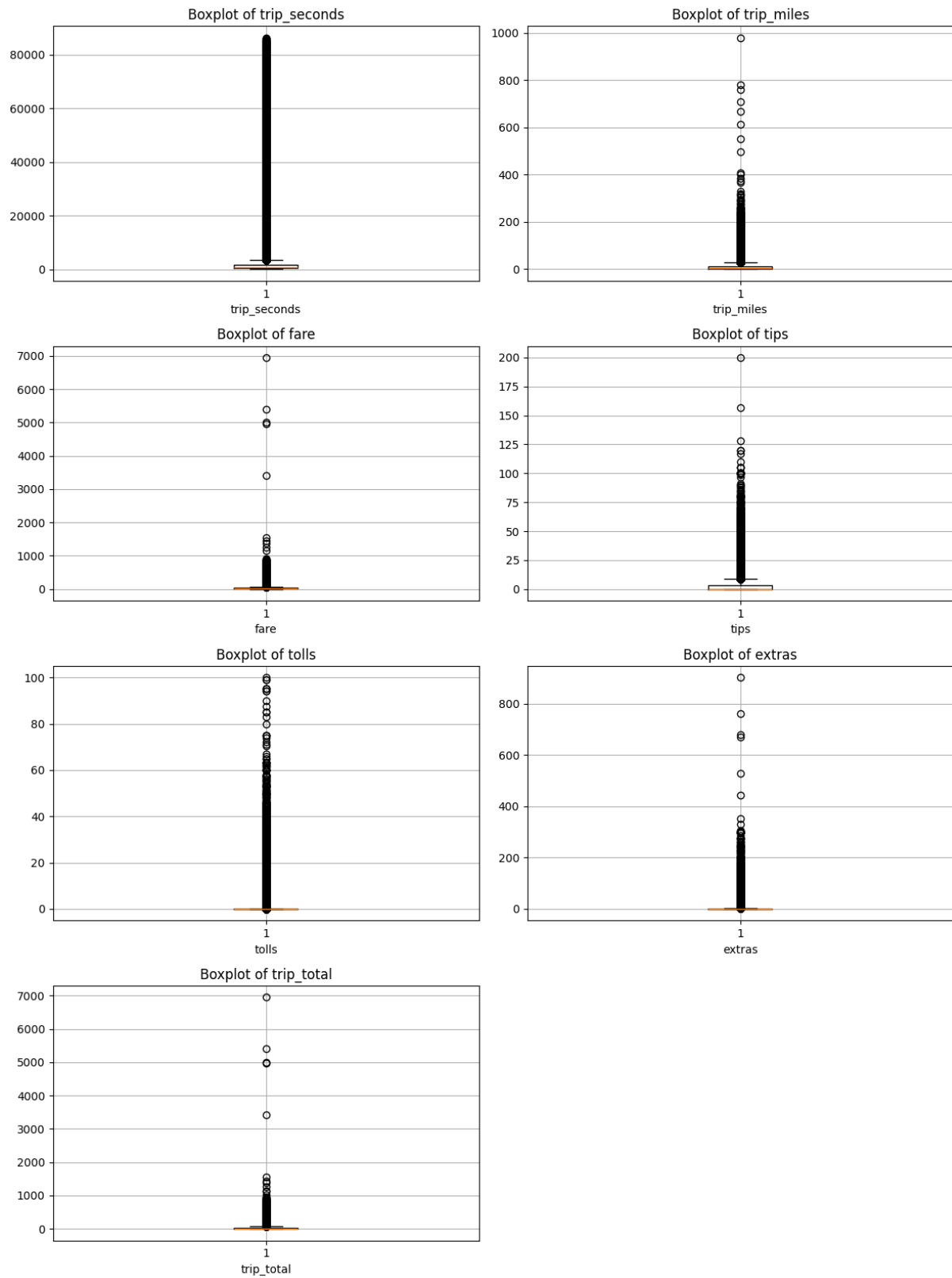


Figure 3: Boxplot of taxi trip attributes to detect outliers

7 Citations and References

You can easily cite your sources using BibLaTeX. For example, you can cite the Quarto documentation (Team, 2024) or a textbook like “R for Data Science” (Wickham et al., 2023). The bibliography will be automatically generated at the end of the document.

References

- City of Chicago. (2024). Taxi trips (2024-) [Accessed: 2026]. https://data.cityofchicago.org/Transportation/Taxi-Trips-2024-/ajtu-isnz/about_data
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